SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05 **Total Marks:50**

Annexure A

Data Interpretation

Criteria	Understanding of Data ***(2.5)***	Analysis ***(2.5)***	Application of Concepts ***(2)***	Conclusion ***(1.5)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I am **Ch.Sathish/(192311032)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch.Sathish

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

ANSWERS

1. Q1. Hospital Patient Segmentation — Elbow Method

cluster have similar characteristics. In this case, the clustering is based on monthly medical expense and health risk score. The appropriate number of clusters can be determined using the elbow K-Means clustering is used to divide patients into groups in which patients within the same method. The given SSE values are:

Number of Clusters (K)	**SSE**
1	600
2	350
3	220
4	200
5	190

SSE (Sum of Squared Errors) measures the total variation within clusters. A lower SSE indicates that patients within each cluster are more similar. However, increasing K will naturally reduce SSE, so we look for the point after which the reduction becomes relatively small.

From K = 1 to K = 2, SSE decreases from 600 to 350, a reduction of 250. From K = 2 to K = 3, SSE decreases from 350 to 220, a reduction of 130. However, from K = 3 to K = 4, the reduction is only 20, and from K = 4 to K = 5, it is only 10.

Therefore, there is a sharp improvement until K = 3, followed by only small improvements. This creates the elbow point at K = 3. Hence, the optimal number of clusters is 3.

The three clusters can be interpreted as different patient segments. One cluster may contain patients with higher health risks, another may contain patients with moderate risk and expenses, and another may contain patients with lower medical requirements. For example, P3 and P4 have relatively high health-risk scores, while P2 and P5 have higher medical expenses but lower risk scores. The actual clusters would depend on the distance calculations performed by KMeans. Selecting three clusters allows the healthcare organization to differentiate patients according to their needs rather than treating all patients identically. High-risk patients can receive closer monitoring and preventive care, while patients with lower risk may be managed through routine follow-ups. The hospital can also use the clusters for resource allocation, treatment planning, insurance analysis, personalized healthcare programs, and budgeting. If too few clusters are selected, patients with significantly different characteristics may be grouped together. On the other hand, selecting too many clusters may produce unnecessarily small groups and make healthcare planning complicated. Thus, K = 3 provides a good balance between simplicity and meaningful patient segmentation.

Conclusion: The elbow point occurs at **K = 3**, making three clusters the most appropriate choice for patient segmentation. This can improve healthcare decision-making by enabling more targeted and efficient patient management.

Q2. E-Commerce Order Analysis — Hierarchical Clustering

Hierarchical clustering, also called HAC (Hierarchical Agglomerative Clustering) when an agglomerative approach is used, creates a hierarchy of groups. Initially, every order is treated as an individual cluster. The algorithm then combines the most similar orders step by step. The resulting hierarchy is represented using a dendrogram. The dataset contains five orders:

Order	Category	Value
O1	Electronics	₹55,000
O2	Clothing	₹25,000
O3	Electronics	₹57,000
O4	Furniture	₹30,000
O5	Clothing	₹27,000

The first important grouping is between O1 and O3, because both belong to Electronics and their order values are very close: ₹55,000 and ₹57,000. Similarly, O2 and O5 are likely to form a cluster because both are Clothing orders with relatively close values of ₹25,000 and ₹27,000.

O4 belongs to Furniture and has an order value of ₹30,000. It is different in category from the other orders, so it may remain separate until a higher dendrogram distance is reached.

A reasonable cut of the dendrogram therefore gives 3 clusters:

- Cluster 1: O1, O3 — Electronics and high-value purchases.
- Cluster 2: O2, O5 — Clothing and medium-value purchases.
- Cluster 3: O4 — Furniture purchase.

The dendrogram is useful because it shows not only the final groups but also how and at what similarity/distance level the groups were merged. If the dendrogram shows a large vertical gap before the final merges, cutting the tree before that large gap is generally appropriate.

These clusters provide useful information about customer purchasing behavior. The electronics cluster indicates customers associated with high-value purchases, which may justify premium offers, warranties, accessories, or product recommendations. The clothing cluster represents customers making relatively lower-value purchases and may respond well to discounts, seasonal offers, and bundled products.

The furniture cluster may represent a distinct customer requirement because furniture purchases can have different buying cycles, pricing, delivery requirements, and promotional strategies.

Hierarchical clustering can therefore help the retailer with customer profiling, personalized recommendations, targeted advertising, inventory management, cross-selling, and promotional campaigns.

Conclusion: A reasonable interpretation of the dendrogram is 3 clusters, consisting mainly of Electronics, Clothing, and Furniture purchasing patterns. These clusters help the retailer understand different customer segments and develop more effective marketing strategies.

Q3. Mobile App User Behavior — Soft/Probability-Based Clustering

The mobile application company wants to segment users based on daily usage time, login frequency, and in-app purchase amount. User behavior is often complex because a person may show characteristics of more than one customer segment. Therefore, probability-based or soft clustering can be useful. In hard clustering, each user is assigned to exactly one cluster. For example, a user would be classified as either an "active user" or a "high-purchase user." In soft clustering, however, each user receives a membership probability for every cluster.

For example, suppose two clusters are identified:

- **Cluster A:** Highly active users
- **Cluster B:** High-purchase users

The membership probabilities could be represented as follows:

User	Cluster A	Cluster B	Interpretation
U1	0.85	0.15	Mainly active
U2	0.20	0.80	Mainly highpurchase
U3	0.95	0.05	Strongly active
U4	0.10	0.90	Strongly highpurchase
U5	0.90	0.10	Mainly active

These probabilities are illustrative;
the actual membership values must be calculated by the clustering algorithm.
From the given data, U1, U3, and U5 have high usage times and high login frequencies. They can therefore be considered highly active users. U2 and U4 have lower usage and login frequency but comparatively high in-app purchases. This suggests a different type of user who may not spend much time in the application but generates significant revenue.

Soft clustering is particularly useful when users are not completely similar to one group. For example, a user who spends a lot of time on the application and also makes frequent purchases could have significant membership in both an activeuser cluster and a high-value customer cluster.

The major advantages of soft clustering are:

- It represents uncertainty in customer classification.
- It identifies users who belong partially to multiple behavioral groups.
- It supports more personalized marketing strategies.

- It can identify users who are transitioning from one behavioral segment to another.
- It provides richer information than simply assigning a single cluster label.
- It can improve recommendation systems and customer-retention strategies.

For example, users with high activity but low purchasing may be targeted with special offers to encourage conversion, while high-purchase users can be offered premium products or loyalty benefits.

Conclusion: Probability-based membership provides a more realistic representation of user behavior because customers do not always fit perfectly into one category. Therefore, soft clustering is valuable for personalization, revenue optimization, customer retention, and behavioral analysis.

Q4. Agricultural Crop Monitoring — Effect of Normalization

Clustering is used in agriculture to identify farms with similar characteristics based on **rainfall, soil fertility, and crop yield**. However, these variables have different numerical scales and units. The dataset is:

Farm	**Rainfall (mm)**	**Soil Fertility**	**Crop Yield**
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Before normalization, rainfall has values ranging from 280 to 520, soil fertility ranges from 3 to 9, and crop yield ranges from 65 to 90. Since clustering algorithms such as K-Means commonly calculate distances between observations, variables with larger numerical scales can have a greater influence on the clustering result.

For example, rainfall has a much larger numerical range than soil fertility. As a result, an unnormalized clustering algorithm may give excessive importance to rainfall. Two farms with similar rainfall but very different soil fertility and yield might be placed in the same cluster.

After normalization, the variables are transformed into a common scale. Techniques such as Min-Max normalization or Z-score standardization can be used. This allows rainfall, soil fertility, and crop yield to contribute more fairly to the distance calculation.

After normalization, the algorithm can identify more meaningful agricultural patterns.

For instance:

- F1 and F4 have high soil fertility and high crop yields.
- F2 and F5 have high rainfall but lower soil fertility and lower crop yields.
- F3 has intermediate values across the variables.

Thus, normalization can change cluster membership because the algorithm is no longer dominated by the variable with the largest numerical scale. Normalization improves clustering by:

15. Giving variables a comparable influence.
16. Preventing large-scale variables from dominating distance calculations.
17. Improving the stability and reliability of cluster formation.
18. Producing clusters that reflect the combined agricultural characteristics.
19. Making the results easier to interpret and compare.
20. Supporting better agricultural decisions regarding irrigation, fertilizer application, crop selection, and farm management.

For example, farms with low fertility and low yield may require soil improvement and fertilizer programs, while farms with high fertility and yield may be suitable for maintaining existing farming practices.

Conclusion: Normalization is important because it prevents differences in measurement scale from distorting the clustering results. It produces more balanced and reliable farm segments and therefore improves agricultural planning, resource allocation, crop management, and decision-making.

Q5. Employee Performance Evaluation — Silhouette Analysis

The company uses clustering to group employees according to productivity, attendance, and skill assessment scores. To determine the best number of clusters, the company uses the silhouette score.

The results are:

Number of Clusters	Silhouette Score
K = 2	0.48
K = 3	**0.68**
K = 4	0.52

The silhouette score evaluates both:

- **Cohesion:** How close an employee is to other employees within the same cluster.
- **Separation:** How far the employee's cluster is from other clusters.

A higher silhouette score indicates that the clusters are more compact internally and better separated from one another. A score close to +1 indicates strong clustering, while a score near zero indicates overlapping clusters.

Among the three models, K = 3 has the highest silhouette score of 0.68. Therefore, the company should select three clusters.

The K = 2 model has a score of 0.48, indicating moderate clustering quality. Increasing the number of clusters to K = 3 improves the score significantly to 0.68. However, increasing K further to 4 reduces the score to 0.52. This indicates that four clusters may divide employees unnecessarily and create groups that are less clearly separated.

Based on the employee scores, the three clusters could conceptually represent:

- **High-performing employees:** High productivity, attendance, and skill scores. E1 and E5 are examples of employees who appear to have strong overall performance.
- **Moderate-performing employees:** Employees with reasonable but not consistently high scores, such as E3.
- **Employees requiring development:** Employees with relatively lower productivity, attendance, and skill scores, such as E2 and E4. The exact cluster membership should be determined by the clustering algorithm rather than assigned only from visual inspection.

The selected three-cluster model can help the organization develop different strategies for different employee groups. High-performing employees can receive recognition, incentives, leadership opportunities, and career advancement. Moderate-performing employees can receive targeted coaching and skill-development opportunities. Employees requiring improvement can be provided with training, mentoring, attendance support, and performance improvement plans.

The model also helps management avoid treating every employee in the same way. Instead, decisions can be based on measurable patterns in productivity, attendance, and skills.

Conclusion: The most suitable model is K = 3 because it has the highest silhouette score of 0.68. It provides the best balance between within-cluster similarity and between-cluster separation and can support more effective employee evaluation, training, recognition, and performance management.